

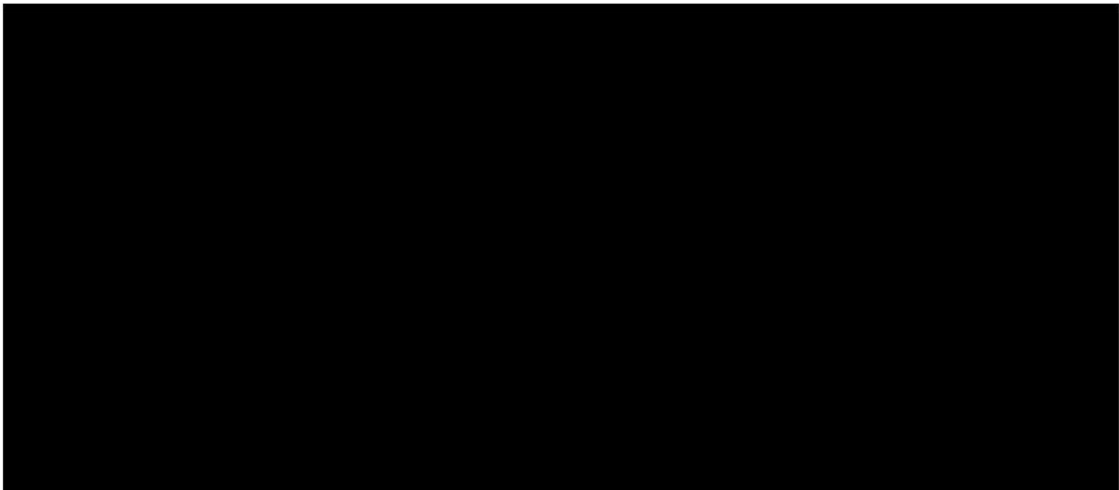
EXHIBIT 27
(Public Version)

Ex. 27C - Domestic Industry Claim Chart
U.S. Patent No. 12,456,334 (“the ’334 patent”)



trinamiX GmbH (“trinamiX”) hereby identifies evidence demonstrating the following trinamiX products practice the ’334 Patent: all trinamiX products that include trinamiX Face Authentication, including but not limited to e.g., the trinamiX Face Authentication software libraries and SDK, ET-90 demonstrator, Cockatiel demonstrator, and MPIC2.0 (collectively, the “Practicing ’334 Products”). These products are made by trinamiX, and are imported, sold for importation, and/or sold within the United States after importation by trinamiX. The chart below is based on evidence of representative products of the Practicing ’334 Products. Each of the following limitations is met literally, and, to the extent a limitation is not met literally, it is met under the doctrine of equivalents.

The trinamiX products identified in this chart are representative of all relevant trinamiX products containing relevant components practicing the patent. The same or similar analysis as discussed below apply to all relevant trinamiX products containing relevant components practicing the patent because, for example, the cameras, sensors, and software operate similarly within each device, and each corresponding device has the same or similar components, features, functions, and/or characteristics. This will further be confirmed through expert discovery.

Claim 1	Practicing ’334 Products
[1pre] A system, comprising:	To the extent the preamble is limiting, the Practicing ’334 Products a system. For example, each of the Practicing ’334 Products is a system that performs face authentication: 

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[1a] a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising an optical sensor, wherein the NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength;	The Practicing '334 Products comprise a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising an optical sensor, wherein the NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength. The Practicing '334 Products comprise a near infrared camera with an optical sensor sensitive to light:

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	A large black rectangular area covering the majority of the table's content, indicating that the information has been redacted.
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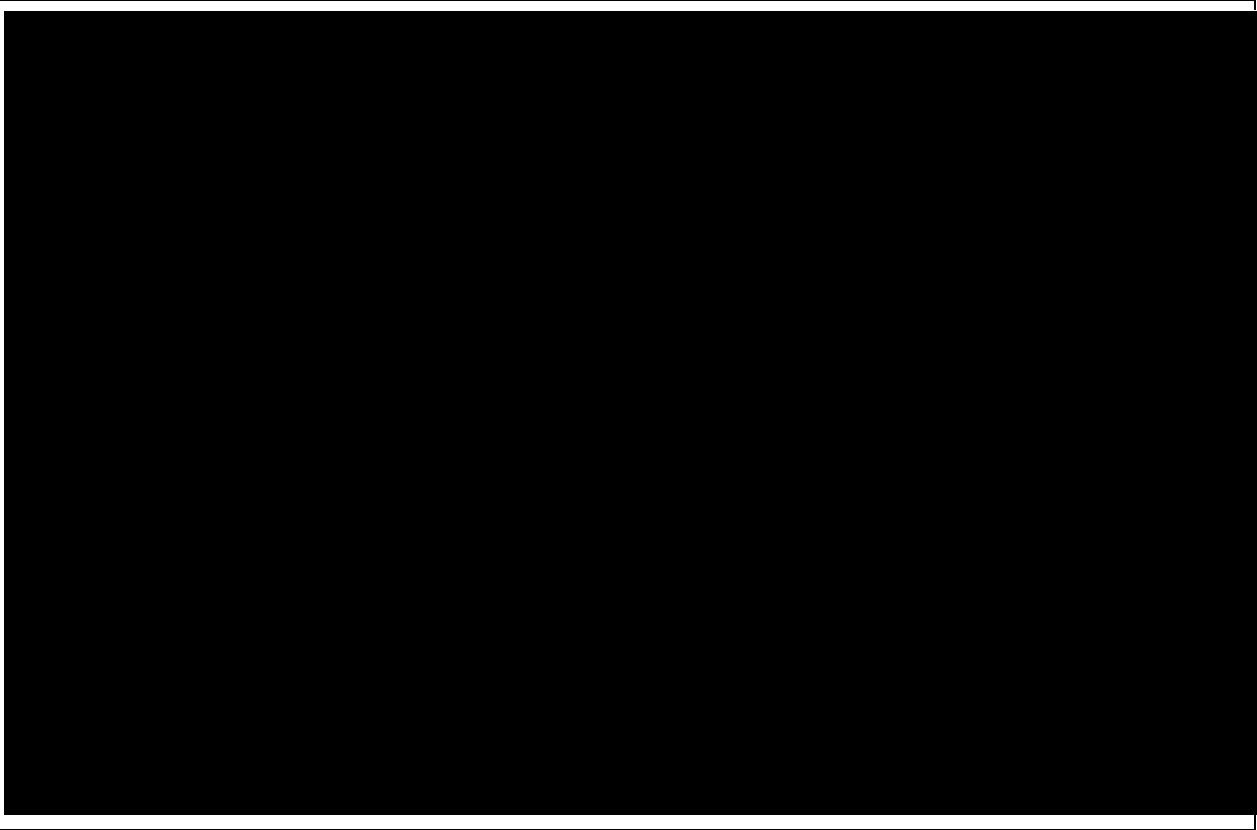
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[1b] a plurality of light projecting devices at least one of which is configured to project a light pattern on the scene, wherein at least one of the plurality of light projecting devices is a vertical cavity surface emitting laser (VCSEL); and	The Practicing '334 Products comprise a plurality of light projecting devices at least one of which is configured to project a light pattern on the scene, wherein at least one of the plurality of light projecting devices is a vertical cavity surface emitting laser (VCSEL). The Practicing '334 Products comprise light projecting devices such a dot projector and flood illuminator that project light patterns:

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	A large black rectangular area covering the entire content of the table, indicating that the information has been redacted.
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	A large black rectangular area used for redaction, covering the majority of the table's content.	
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	A large black rectangular area used to redact the main body of the claim chart.
[1c] a processor configured to:	The Practicing '334 Products comprise a processor. The Practicing '334 Products comprise a processor unit, such as a Qualcomm HDK8450 chip processor:

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	A large black rectangular area used for redaction, covering the majority of the table's content.	
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[1d] output a first image comprising light reflected from the scene in response to the projection by the light projecting devices and output a second image comprising a reflective light pattern reflected from the scene in response to the projection by the light projecting devices;	The Practicing ’334 Products comprise a processor configured to output a first image comprising light reflected from the scene in response to the projection by the light projecting devices and output a second image comprising a reflective light pattern reflected from the scene in response to the projection by the light projecting devices. The Practicing ’334 Products comprise a processor (see 1[c]) and outputs images with reflective light patterns from its light projecting devices such a dot projector and flood illuminator (see 1[b]) that perform this claim limitation.

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	A large black rectangular area used for redaction, covering the majority of the top half of the table.
[1e] detect a face in a region in the first image;	The Practicing '334 Products comprise a processor configured to detect a face in a region in the first image.

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	A large solid black rectangular area used to redact the content of the first row of the table.
[1f] determine a distribution of light intensity of a region in the second image;	The Practicing '334 Products comprise a processor configured to determine a distribution of light intensity of a region in the second image.

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	A large black rectangular area used for redaction, covering the top half of the table's content area.
[1g] compare the determined distribution of light intensity to a distribution of light intensity derived from a reference image;	The Practicing '334 Products comprise a processor configured to compare the determined distribution of light intensity to a distribution of light intensity derived from a reference image.

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[1h] determine a material classification of the face based on the comparison of the determined distribution of light intensity and the distribution of light intensity derived from the reference image;	The Practicing '334 Products comprise a processor configured to determine a material classification of the face based on the comparison of the determined distribution of light intensity and the distribution of light intensity derived from the reference image.

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	A large solid black rectangular area used for redaction, covering the upper right portion of the table.
[1i] determine at least one depth map based on depth differences in the reflective light pattern; and	The Practicing '334 Products comprise a processor configured to determine at least one depth map based on depth differences in the reflective light pattern.

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	A large solid black rectangular area used for redaction, covering the top half of the table's content area.
[1j] authenticate the face based on the detection of the face, the classification of the face, and the depth map.	The Practicing '334 Products comprise a processor configured to authenticate the face based on the detection of the face, the classification of the face, and the depth map.

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